AP Calculus BC
Unit 3 Review Book

Differentiation: Composite, Implicit, and Inverse Functions

Purpose. This review book is written for a student who has already finished the course and now needs a full, beginner-friendly, exam-focused review. It is intentionally explicit: ideas are unpacked slowly, formulas are highlighted clearly, and common AP mistakes are addressed directly.

What this unit really tests

- The **Chain Rule** for composite functions.
- **Implicit differentiation**, especially when x and y are mixed together.
- **Inverse function derivatives**, including the standard inverse trigonometric derivatives.
- Choosing the **right differentiation tool** on mixed expressions.
- Computing and interpreting **higher-order derivatives**.

Prepared as a detailed final-review reference with fully rendered mathematics.

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1. Unit Map and Core Ideas

The big picture. Unit 3 adds a new layer to derivative work. In Unit 2, many derivatives were direct: power rule, product rule, quotient rule, and basic special-function derivatives. In Unit 3, the main challenge is structure. A function may be nested inside another function, hidden inside an equation with x and y, or reversed as an inverse function. The derivative rules are still the same, but now you must recognize the structure first.

1.1 Essential formulas you should know before the chapter details

$$\frac{d}{dx}[f(g(x))] = f'(g(x))\,g'(x)$$

$$\frac{d}{dx}(y^n) = ny^{n-1}\frac{dy}{dx}$$

$$(f^{-1})'(a) = \frac{1}{f'(f^{-1}(a))}$$

$$\frac{d}{dx}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}, \qquad \frac{d}{dx}(\arccos x) = -\frac{1}{\sqrt{1-x^2}}, \qquad \frac{d}{dx}(\arctan x) = \frac{1}{1+x^2}$$

$$f''(x) = \frac{d}{dx}(f'(x))$$

Most common Unit 3 mistake: a student sees a complicated function and differentiates the "outside" but forgets to multiply by the derivative of the "inside." That one missing factor causes a large amount of avoidable point loss.

2. 3.1 The Chain Rule

2.1 What the Chain Rule means

A composite function has the form

$$y = f(g(x)).$$

There is an *inside function* $g(x)$ and an *outside function* $f(\cdot)$. The Chain Rule says:

$$\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x).$$

In words:

> Differentiate the outside function, keep the inside expression unchanged, and then multiply by the derivative of the inside.

2.2 Core pattern templates

$$\frac{d}{dx}[(u(x))^n] = n(u(x))^{n-1}u'(x)$$

$$\frac{d}{dx}[\sin(u(x))] = \cos(u(x))u'(x)$$

$$\frac{d}{dx}[\cos(u(x))] = -\sin(u(x))u'(x)$$

$$\frac{d}{dx}[e^{u(x)}] = e^{u(x)}u'(x)$$

$$\frac{d}{dx}[\ln(u(x))] = \frac{u'(x)}{u(x)} \qquad \frac{d}{dx}[\sqrt{u(x)}] = \frac{u'(x)}{2\sqrt{u(x)}}$$

2.3 How to recognize a composite function quickly

If one function is *plugged into* another, you probably need the Chain Rule. Typical AP forms are

$$(3x^2-1)^5, \qquad \sin(x^3), \qquad e^{2x+1}, \qquad \ln(4-x), \qquad \sqrt{1+x^4}.$$

Example 1. Differentiate

$$y = (2x^3-1)^4.$$

Solution. Let $u = 2x^3 - 1$. Then

$$y = u^4 \implies \frac{dy}{dx} = 4u^3\frac{du}{dx}.$$

Since

$$\frac{du}{dx} = 6x^2,$$

we get

$$y' = 4(2x^3-1)^3(6x^2) = 24x^2(2x^3-1)^3.$$

Example 2. Differentiate

$$y = \sin(5x^2+1).$$

Solution. The outside function is $\sin(\cdot)$ and the inside function is $5x^2+1$.

$$y' = \cos(5x^2+1)\cdot\frac{d}{dx}(5x^2+1) = 10x\cos(5x^2+1).$$

Example 3. Differentiate

$$y = \ln(3x^2-4x+7).$$

Solution.

$$y' = \frac{6x-4}{3x^2-4x+7}.$$

Notice the structure: derivative of $\ln(u)$ is always u'/u.

2.4 Typical AP multiple-choice question

Differentiate

$$f(x) = e^{x^2}\cos x.$$

This is not only a Chain Rule problem. It also needs the Product Rule:

$$\begin{aligned} f'(x) &= \left(e^{x^2}\right)'\cos x + e^{x^2}(\cos x)' \\ &= \left(e^{x^2}\cdot 2x\right)\cos x - e^{x^2}\sin x \\ &= e^{x^2}(2x\cos x - \sin x). \end{aligned}$$

This is the exact kind of mixed-structure question AP likes.

Chain Rule danger signs.

- Writing $\frac{d}{dx}[\sin(x^2)]$ as $\cos(x^2)$ instead of $2x\cos(x^2)$.
- Writing $\frac{d}{dx}[\ln(u)]$ as $1/\ln(u)$ instead of u'/u.
- Forgetting that nested radicals and exponentials are also composite functions.

3. 3.2 Implicit Differentiation

3.1 Why implicit differentiation exists

Sometimes a relation is not solved for y. For example,

$$x^2 + y^2 = 25.$$

This equation describes a curve, but it is not written as $y = f(x)$. Implicit differentiation lets us differentiate both sides with respect to x without solving first.

3.2 The key rule you must remember

Whenever you differentiate a term containing y, you must treat y as a function of x. That means:

$$\frac{d}{dx}(y) = \frac{dy}{dx}, \qquad \frac{d}{dx}(y^n) = ny^{n-1}\frac{dy}{dx}.$$

More examples:

$$\frac{d}{dx}(\sin y) = \cos y\frac{dy}{dx}, \qquad \frac{d}{dx}(e^y) = e^y\frac{dy}{dx}, \qquad \frac{d}{dx}(\ln y) = \frac{1}{y}\frac{dy}{dx}.$$

3.3 Standard workflow

1. Differentiate both sides with respect to x.
2. Every time y appears, attach a factor of $\frac{dy}{dx}$.
3. Collect all $\frac{dy}{dx}$ terms on one side.
4. Factor out $\frac{dy}{dx}$.
5. Solve.

Example 4. Given

$$x^2 + y^2 = 10,$$

find $\frac{dy}{dx}$.
Differentiate both sides:

$$2x + 2y\frac{dy}{dx} = 0.$$

Solve for $\frac{dy}{dx}$:

$$2y\frac{dy}{dx} = -2x$$
$$\frac{dy}{dx} = -\frac{x}{y}.$$

Example 5. Given

$$x^2 + xy + y^2 = 7,$$

find $\frac{dy}{dx}$.
Differentiate term by term:

$$2x + \frac{d}{dx}(xy) + 2y\frac{dy}{dx} = 0.$$

Use the Product Rule on xy:

$$\frac{d}{dx}(xy) = x\frac{dy}{dx} + y.$$

So

$$2x + x\frac{dy}{dx} + y + 2y\frac{dy}{dx} = 0.$$

Group derivative terms:

$$(x + 2y)\frac{dy}{dx} = -(2x + y).$$

Therefore,

$$\frac{dy}{dx} = \frac{-(2x + y)}{x + 2y}.$$

Example 6. Find the slope of the tangent line to

$$x^2 + y^2 = 25$$

at the point $(3, 4)$.
From the earlier result,

$$\frac{dy}{dx} = -\frac{x}{y}.$$

Substitute $(3, 4)$:

$$\left.\frac{dy}{dx}\right|_{(3,4)} = -\frac{3}{4}.$$

So the tangent-line equation is

$$y - 4 = -\frac{3}{4}(x - 3).$$

3.4 AP-style extension

If the relation contains trigonometric or exponential terms, the same rule still applies.

Example 7. Given

$$x + \sin y = y^2,$$

find $\frac{dy}{dx}$.
Differentiate:

$$1 + \cos y \frac{dy}{dx} = 2y \frac{dy}{dx}.$$

Collect derivative terms:

$$1 = (2y - \cos y)\frac{dy}{dx}.$$

Thus,

$$\frac{dy}{dx} = \frac{1}{2y - \cos y}.$$

Most common implicit-differentiation mistakes.

- Forgetting the extra factor $\frac{dy}{dx}$ after differentiating a y-term.
- Forgetting the Product Rule on xy, x^2y, or similar terms.
- Plugging in the point before finishing the differentiation.

4. 3.3 Differentiating Inverse Functions

4.1 Main formula

If f is differentiable and invertible, then the derivative of its inverse is

$$(f^{-1})'(a) = \frac{1}{f'(f^{-1}(a))}.$$

A very common AP version is: if $f(b) = a$, then

$$(f^{-1})'(a) = \frac{1}{f'(b)}.$$

4.2 Interpretation

This formula says: to differentiate the inverse at an output value, you go back to the corresponding input value on the original function, evaluate the original derivative there, and then take the reciprocal.

Example 8. Suppose

$$f(2) = 5, \qquad f'(2) = 4.$$

Find

$$(f^{-1})'(5).$$

Because $f(2) = 5$, the corresponding input is 2. Therefore,

$$(f^{-1})'(5) = \frac{1}{f'(2)} = \frac{1}{4}.$$

Example 9. If

$$f(x) = x^3 + 1,$$

find $(f^{-1})'(9)$.
First solve $f(x) = 9$:

$$x^3 + 1 = 9 \implies x^3 = 8 \implies x = 2.$$

Now

$$f'(x) = 3x^2, \qquad f'(2) = 12.$$

Therefore,

$$(f^{-1})'(9) = \frac{1}{12}.$$

4.3 Why inverse derivatives matter on AP

AP often gives a table of values and asks for an inverse derivative without asking you to find the inverse explicitly. This is meant to test whether you understand the relationship between f and f^{-1}.

Classic AP trap. If the question asks for $(f^{-1})'(7)$ and the table says $f(3) = 7$, then you must use $f'(3)$, not $f'(7)$. The input for the derivative comes from the original function, not from the inverse notation alone.

5. 3.4 Differentiating Inverse Trigonometric Functions

5.1 Must-memorize formulas

$$\frac{d}{dx}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}$$
$$\frac{d}{dx}(\arccos x) = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\arctan x) = \frac{1}{1+x^2}$$

Optional BC familiarity beyond the main three:

$$\frac{d}{dx}(\operatorname{arcsec} x) = \frac{1}{|x|\sqrt{x^2-1}}, \qquad \frac{d}{dx}(\operatorname{arccsc} x) = -\frac{1}{|x|\sqrt{x^2-1}}.$$

5.2 Chain Rule versions

If the inside is not just x, multiply by the inside derivative:

$$\frac{d}{dx}(\arcsin(u(x))) = \frac{u'(x)}{\sqrt{1-u(x)^2}}$$

$$\frac{d}{dx}(\arccos(u(x))) = -\frac{u'(x)}{\sqrt{1-u(x)^2}}$$

$$\frac{d}{dx}(\arctan(u(x))) = \frac{u'(x)}{1+u(x)^2}$$

Example 10. Differentiate

$$y = \arctan(3x).$$

Using the Chain Rule,

$$y' = \frac{3}{1+(3x)^2} = \frac{3}{1+9x^2}.$$

Example 11. Differentiate

$$y = \arcsin(x^2).$$

Then

$$y' = \frac{2x}{\sqrt{1-x^4}}.$$

Example 12. Differentiate

$$y = \arccos(1-2x).$$

Then

$$y' = -\frac{-2}{\sqrt{1-(1-2x)^2}} = \frac{2}{\sqrt{1-(1-2x)^2}}.$$

Inverse trig warning list.

- $\arcsin x$ is not the same as $1/\sin x$.
- $\arccos x$ has a negative derivative; that sign is frequently tested.
- If the inside is not x, Chain Rule is mandatory.

6. 3.5 Selecting Procedures for Calculating Derivatives

6.1 The real skill in this section

This section is about choosing the correct combination of rules. Many AP questions in Unit 3 are not about one rule in isolation. Instead, they ask whether you can recognize the structure and choose an efficient path.

6.2 Decision guide

Quick selection guide

- **Nested expression?** Use the Chain Rule.
- **Product of two functions?** Use the Product Rule.
- **Quotient of two functions?** Use the Quotient Rule.
- **x and y mixed in one equation?** Use implicit differentiation.
- **Inverse function or inverse trig?** Use the inverse derivative formulas.

Example 13. Differentiate

$$y = (x^2+1)^5 \cos x.$$

This is a product, and one factor is composite. So use Product Rule first, then Chain Rule:

$$y' = \left[5(x^2+1)^4(2x)\right]\cos x + (x^2+1)^5(-\sin x).$$

Thus,

$$y' = 10x(x^2+1)^4\cos x - (x^2+1)^5 \sin x.$$

Example 14. Differentiate

$$y = \frac{\ln x}{x^2+1}.$$

This is a quotient, and the numerator itself is a special derivative. Using the Quotient Rule,

$$y' = \frac{\left(\frac{1}{x}\right)(x^2+1) - \ln x(2x)}{(x^2+1)^2}.$$

Example 15. Differentiate

$$y = e^{\sin x}.$$

This is composite twice: exponential outside, sine inside.

$$y' = e^{\sin x} \cdot \cos x.$$

6.3 How AP often tests this section

AP frequently asks a question where the main challenge is not the algebra but identifying the structure. Students who slow down for five seconds to label the function as *product*, *quotient*, *implicit*, or *inverse* usually do much better than students who start differentiating immediately.

7. 3.6 Calculating Higher-Order Derivatives

7.1 Definitions

The second derivative is

$$f''(x) = \frac{d}{dx}(f'(x)).$$

The third derivative is

$$f^{(3)}(x) = \frac{d}{dx}(f''(x)).$$

And so on.

If $y = f(x)$, you may also see

$$y'', \qquad y^{(3)}, \qquad \frac{d^2y}{dx^2}, \qquad \frac{d^3y}{dx^3}.$$

7.2 Why higher-order derivatives matter

In later units, second derivatives are used for concavity, acceleration, Taylor polynomials, and more. In Unit 3, the goal is simply to compute them accurately, including in implicit settings.

Example 16. If

$$f(x) = x^3 - 3x^2 + 2,$$

find $f'(x)$ and $f''(x)$.
First derivative:

$$f'(x) = 3x^2 - 6x.$$

Second derivative:

$$f''(x) = 6x - 6.$$

Example 17. If

$$y = x^2 \sin x,$$

find y''.
First derivative using Product Rule:

$$y' = 2x \sin x + x^2 \cos x.$$

Differentiate again:

$$y'' = 2 \sin x + 2x \cos x + 2x \cos x - x^2 \sin x.$$

So

$$y'' = 2 \sin x + 4x \cos x - x^2 \sin x.$$

7.3 Higher-order derivatives with implicit functions

These are possible but usually more technical. The usual method is:

1. Find $\frac{dy}{dx}$ first.
2. Differentiate that result again with respect to x.
3. Substitute the original relation if simplification is needed.

Example 18. For the circle

$$x^2 + y^2 = 25,$$

we already found

$$\frac{dy}{dx} = -\frac{x}{y}.$$

Differentiate again using the Quotient Rule:

$$y'' = \frac{d}{dx}\left(-\frac{x}{y}\right) = -\frac{y - x\frac{dy}{dx}}{y^2}.$$

Now substitute $\frac{dy}{dx} = -\frac{x}{y}$:

$$y'' = -\frac{y - x\left(-\frac{x}{y}\right)}{y^2} = -\frac{y + \frac{x^2}{y}}{y^2} = -\frac{x^2 + y^2}{y^3}.$$

Since $x^2 + y^2 = 25$,

$$y'' = -\frac{25}{y^3}.$$

Higher-order derivative warning. Errors compound quickly. If the first derivative is not simplified or is already incorrect, the second derivative usually becomes much harder than it needs to be. Always pause and clean up $f'(x)$ before differentiating again.

8. Full Unit 3 Formula Sheet

Core formulas to memorize

$$\frac{d}{dx}[f(g(x))] = f'(g(x))g'(x)$$

$$\frac{d}{dx}(y^n) = ny^{n-1}\frac{dy}{dx}$$

$$(f^{-1})'(a) = \frac{1}{f'(f^{-1}(a))}$$

$$\frac{d}{dx}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\arccos x) = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\arctan x) = \frac{1}{1+x^2}$$

$$f''(x) = \frac{d}{dx}(f'(x))$$

9. Exam Strategy for Unit 3

9.1 A practical decision sequence

When you see a derivative question in Unit 3, use this checklist:

1. Is the function written explicitly, or is it implicit?
2. Is there a nested expression? If yes, use Chain Rule.
3. Is there a product or quotient outside that nested expression?
4. Is the problem really asking for an inverse derivative?
5. If it is an inverse trig function, do you know the exact derivative formula from memory?

9.2 The three biggest AP habits to build

- **Label the structure before differentiating.**
- **Carry $\frac{dy}{dx}$ every time y appears in an implicit problem.**
- **For inverse derivatives, first find the matching input on the original function.**

10. Mixed Review Problems with Fully Worked Solutions

Mixed Problem 1. Differentiate

$$y = \ln(\sqrt{1+x^4}).$$

Treat the square root as the inside of the logarithm:

$$y' = \frac{\frac{d}{dx}\left(\sqrt{1+x^4}\right)}{\sqrt{1+x^4}}.$$

Now

$$\frac{d}{dx}\left(\sqrt{1+x^4}\right) = \frac{1}{2\sqrt{1+x^4}} \cdot 4x^3 = \frac{2x^3}{\sqrt{1+x^4}}.$$

Therefore,

$$y' = \frac{2x^3}{1+x^4}.$$

Mixed Problem 2. Find $\frac{dy}{dx}$ if

$$x^3 + y^3 = 6xy.$$

Differentiate both sides:

$$3x^2 + 3y^2\frac{dy}{dx} = 6\left(x\frac{dy}{dx} + y\right).$$

Expand the right side:

$$3x^2 + 3y^2\frac{dy}{dx} = 6x\frac{dy}{dx} + 6y.$$

Move derivative terms together:

$$3y^2\frac{dy}{dx} - 6x\frac{dy}{dx} = 6y - 3x^2.$$

Factor:

$$(3y^2 - 6x)\frac{dy}{dx} = 6y - 3x^2.$$

So

$$\frac{dy}{dx} = \frac{6y - 3x^2}{3y^2 - 6x} = \frac{2y - x^2}{y^2 - 2x}.$$

Mixed Problem 3. Suppose $f(1) = 4$ and $f'(1) = -3$. Find $(f^{-1})'(4)$.
Because $f(1) = 4$, we use the reciprocal of the original derivative at $x = 1$:

$$(f^{-1})'(4) = \frac{1}{f'(1)} = -\frac{1}{3}.$$

Mixed Problem 4. Differentiate

$$y = \arcsin\left(\frac{x}{2}\right).$$

Using the Chain Rule,

$$y' = \frac{\frac{1}{2}}{\sqrt{1-\left(\frac{x}{2}\right)^2}} = \frac{1}{2\sqrt{1-\frac{x^2}{4}}}.$$

A simplified equivalent form is also acceptable.

Mixed Problem 5. Find the second derivative of

$$f(x) = e^{x^2}.$$

First derivative:

$$f'(x) = 2xe^{x^2}.$$

Differentiate again using Product Rule:

$$f''(x) = 2e^{x^2} + 2x\left(2xe^{x^2}\right)$$

$$= 2e^{x^2} + 4x^2e^{x^2} = 2e^{x^2}(1+2x^2).$$

11. Final Memory Checklist

Before the AP exam, you should be able to do each of the following without hesitation:

- Differentiate a nested expression using the Chain Rule.
- Differentiate equations where x and y are mixed together.
- Find slopes of tangent lines on implicit curves.
- Use the inverse-function derivative formula from a table or function equation.
- Recall the derivatives of $\arcsin x$, $\arccos x$, and $\arctan x$ instantly.
- Compute second derivatives, including on expressions that already used Product Rule or Chain Rule once.
- Identify the correct derivative method before beginning the algebra.